SYED MOHAMMAD ADEEL

AI / ML Engineer · Backend & Full-Stack Developer

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SUMMARY

Computer Science graduate (VIT-AP, 2026) who builds and ships AI-powered backend systems in Python. Author of a peer-reviewed paper on a multi-agent LLM search engine, accepted for oral presentation at CISCom 2026 (University of Malta) and to appear in the Springer CCIS series. Hands-on with LLM applications — multi-agent orchestration, embeddings, vector search and retrieval (RAG) — plus computer-vision model training (YOLO, 86% mAP@0.5) and production practice: REST API design, Redis caching, and CI-tested deployment on AWS. Cut API error/fallback rate 24% on an AI search service through structured testing and prompt refinement.

PUBLICATION

"SEFGH: A Multi-Agent LLM-Based Search Engine for GitHub Repositories" — 2nd International Conference on Computational Intelligence and Soft Computing (CISCom 2026), L-Università ta' Malta, Msida, Malta, 14–16 October 2026. **Accepted for oral presentation; to appear in Springer CCIS** (indexed in Scopus, INSPEC, zbMATH and SCImago).

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, SQL, C/C++

AI / ML: Multi-agent LLM systems, embeddings & vector search, RAG pipelines, LLM API integration, prompt engineering & evaluation, PyTorch, TensorFlow, scikit-learn, OpenCV, YOLOv5/v7

Backend & APIs: FastAPI, Flask, Node/Express, REST API design, Redis caching, Celery (async task queues), rate limiting & pagination

Frontend: React.js, Next.js, TypeScript, Tailwind CSS

Databases: PostgreSQL, MySQL, MongoDB, Redis

Cloud & DevOps: AWS (EC2, S3, Lambda), Docker, Git, GitHub Actions (CI/CD), Linux

Testing & Quality: Automated test suites, CI-based testing, structured test-case design, debugging, data logging

PROJECTS

SEFGH — Multi-Agent LLM Search Engine for GitHub (peer-reviewed) *Python · FastAPI · LLM Agents · Redis*

- Designed a multi-agent LLM architecture in which specialised agents decompose a natural-language query, generate and refine optimised GitHub search expressions, and rank the returned repositories — the system published as a peer-reviewed conference paper.
- Built the full-stack service around the GitHub REST API with rate limiting, Redis caching and server-side pagination; implemented semantic result ranking with embeddings and vector search.
- Added structured logging and a CI test suite (GitHub Actions) covering query parsing, agent routing, ranking and API fallbacks — reducing response errors/fallbacks by 24% through structured test-case development and iterative prompt refinement.

Scalable Palm-Tree Detection Service *Computer Vision · YOLOv5/v7 · Flask · Celery · PostgreSQL · AWS S3*

- Trained and compared YOLOv5 and YOLOv7 object-detection models on aerial imagery, reaching 86% mAP@0.5, and served inference through a Flask REST API.
- Engineered asynchronous batch processing with Celery so large image sets are handled without blocking the API.
- Designed the PostgreSQL schema and ingestion pipeline with Amazon S3 image storage for a scalable end-to-end pipeline.

Autonomous Tiny Drone — Flight Controller & Test Harness *Embedded C/C++ · Arduino/ESP32 · Python · UART*

- Wrote embedded C/C++ firmware implementing PID control loops for autonomous flight stabilisation with timing-critical sensor integration.
- Built a Python telemetry ground station over serial (UART) with live monitoring, plus automated test scripts validating stabilisation and sensor readings across flight scenarios.

EDUCATION

Vellore Institute of Technology – AP 2022 – 2026

B.Tech in Computer Science and Engineering CGPA: 8.14 / 10

CERTIFICATIONS & ACTIVITIES

- NVIDIA Deep Learning Institute — Fundamentals of Deep Learning.